# Outline

1 Modular Arithmetic Tools

2 Introduction to Coding Theory

## 57. Explanation

*Main idea of the Coding Theory: to fluff the information so that meaningful vectors are "far" from each other. Ideas of detecting/correcting capacity and cost.*

## 58. Definition

*Let $A$ be an alphabet of $q \geqslant 2$ symbols (when $q = 2$ we say it is the binary alphabet). A (bloc) code is a non-empty subset $\emptyset \neq \mathcal{C} \subseteq A^n$.*

## 59. Definition

*The Hamming distance is the map $d : A^n \to A^n$ defined as*
$d(\mathbf{x}, \mathbf{y}) = d((x_1, \ldots, x_n), (y_1, \ldots, y_n)) = |\{1 \leqslant i \leqslant n \mid x_i \neq y_i\}|$.

## 60. Observation

*The Hamming distance has the following properties:*
*(i)* $d(\mathbf{x}, \mathbf{y}) \geqslant 0$ *and* $d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$;
*(ii)* $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$;
*(iii)* $d(\mathbf{x}, \mathbf{y}) \leqslant d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$.

## 61. Definition

*Let $\mathcal{C} \subseteq A^n$ be a code. The length of $\mathcal{C}$ is $n$; the size of $\mathcal{C}$ is $M = |\mathcal{C}| \leqslant q^n$; the distance of $\mathcal{C}$ is $\delta(\mathcal{C}) = \min\{d(\mathbf{u}, \mathbf{v}) \mid \mathbf{u}, \mathbf{v} \in \mathcal{C}, \mathbf{u} \neq \mathbf{v}\}$. $\mathcal{C}$ is an $(n, M, \delta)_q$-code.*

## 62. Example

*The trivial code, the total code, the repetition code.*

## 63. Observation

*Let $\mathcal{C} \subseteq A^n$ be a binary code and let $\overline{\mathcal{C}} = \{(u_1, \ldots, u_n, u_{n+1}) \mid (u_1, \ldots, u_n) \in \mathcal{C}$ and $u_{n+1}$ is a parity bit$\}$. Then $\overline{\mathcal{C}}$ is an $(n+1, M, \overline{\delta})_2$-code, where $\overline{\delta}$ is $\delta$ or $\delta + 1$ depending on whether $\delta$ is even or odd, resp.*

### 2.2. Balls, radius of a code, perfect codes

## 64. Definition

*The ball of center $\mathbf{x} \in A^n$ and radius $r$ is the set $B(\mathbf{x}, r) = \{\mathbf{y} \in A^n \mid d(\mathbf{x}, \mathbf{y}) \leqslant r\}$.*

## 65. Observation

$|B(\mathbf{x}, r)| = \binom{n}{0}(q-1)^0 + \binom{n}{1}(q-1)^1 + \cdots + \binom{n}{r}(q-1)^r$.

## 66. Example

## 67. Observation

*Let* $\mathbf{x}, \mathbf{y} \in A^n$. *If* $d(\mathbf{x}, \mathbf{y}) = r$ *and* $1 \leqslant s \leqslant r \leqslant n$, *then there exists* $\mathbf{z} \in A^n$ *such that* $d(\mathbf{x}, \mathbf{z}) = s$ *and* $d(\mathbf{z}, \mathbf{y}) = r - s$.

## 68. Definition

*Let* $\mathcal{C} \subseteq A^n$ *be a code. The tangent radius and the covering radius of* $\mathcal{C}$ *are, resp.,*

$$\rho(\mathcal{C}) = \max\{r \mid B(\mathbf{u}, r) \cap B(\mathbf{v}, r) = \emptyset,\ \forall \mathbf{u} \neq \mathbf{v} \in \mathcal{C}\};$$
$$\tau(\mathcal{C}) = \min\{r \mid \cup_{\mathbf{u} \in \mathcal{C}} B(\mathbf{u}, r) = A^n\}.$$

## 69. Example

## 70. Proposition

*Let* $\mathcal{C}$ *be a* $(n, M, \delta)_q$*-code. Then,* $\rho(\mathcal{C}) = \lfloor \frac{\delta - 1}{2} \rfloor$ *and* $\tau(\mathcal{C}) = \max\{d(\mathbf{x}, \mathcal{C}) \mid \mid \mathbf{x} \in A^n\}$, *where* $d(\mathbf{x}, \mathcal{C}) = \min\{d(\mathbf{x}, \mathbf{u}) \mid \mathbf{u} \in \mathcal{C}\}$.

## 71. Definition

*An $(n, M, \delta)_q$-code $\mathcal{C}$ is perfect if $\rho(\mathcal{C}) = \tau(\mathcal{C})$; equivalently, if $M \sum_{i=0}^{\rho(\mathcal{C})} \binom{n}{i}(q-1)^i = q^n$, or if $M \sum_{i=0}^{\tau(\mathcal{C})} \binom{n}{i}(q-1)^i = q^n$.*

## 72. Example

## 73. Proposition

*If $\mathcal{C}$ is perfect then $\delta(\mathcal{C})$ is odd. The converse is not true.*

### 2.3. Error detection and error correction capabilities

## 74. Definition

*Let $\mathcal{C}$ be a code. We say that $\mathcal{C}$ detects $s$ errors if $\forall \mathbf{u} \in \mathcal{C}$, $B(\mathbf{u}, s) \cap \mathcal{C} = \{\mathbf{u}\}$. We say that $\mathcal{C}$ corrects $t$ errors if $\forall \mathbf{u} \neq \mathbf{v} \in \mathcal{C}$, $\forall \mathbf{x} \in A^n$, we have that $d(\mathbf{x}, \mathbf{u}) \leqslant t \Rightarrow d(\mathbf{x}, \mathbf{u}) < d(\mathbf{x}, \mathbf{v})$.*

## 75. Observation

*An $(n, M, \delta)_q$-code detects up to $\delta - 1$ errors, and corrects up to $\lfloor \frac{\delta-1}{2} \rfloor$ errors.*

## 2.4. The fundamental problem of coding theory

### 76. Explanation

*Contradictory interests: we want codes with $n$ 'small', $M$ 'large', and $\delta$ 'large'.*

### 77. Definition

*The fundamental problem of coding theory consists on computing the function* $A_q(n,\delta) = \max\{M \mid \exists \text{ an } (n,M,\delta)_q - \text{code}\}$, *for* $q \geqslant 2$, $0 \leqslant \delta \leqslant n$. *An* $(n,M,\delta)_q$*-code is optimal if* $M = A_q(n,\delta)$.

### 78. Example

$A_q(n,0) = 1$, $A_q(n,1) = q^n$, $A_q(n,n) = q$, $A_2(4,3) = 2$. *Table of known values.*

### 79. Proposition

*Let* $n \geqslant \delta \geqslant 1$. *Then,*
*(i)* $A_q(n+1,\delta+1) \leqslant A_q(n,\delta)$;
*(ii) if* $\delta$ *is even then* $A_2(n,\delta) = A_2(n-1,\delta-1)$.

## 80. Theorem (Hamming bound)

*Let* $n \geqslant \delta \geqslant 1$ *and write* $\rho = \lfloor \frac{\delta-1}{2} \rfloor$. *Then,* $A_q(n,\delta) \leqslant \frac{q^n}{\sum_{i=0}^{\rho} \binom{n}{i}(q-1)^i}$.

## 81. Theorem (Gilbert–Varshamov bound)

*Let* $n \geqslant \delta \geqslant 1$. *Then,* $\frac{q^n}{\sum_{i=0}^{\delta-1} \binom{n}{i}(q-1)^i} \leqslant A_q(n,\delta)$.

## 82. Theorem (Singleton bound)

*Let* $n \geqslant \delta \geqslant 1$. *Then,* $A_q(n,\delta) \leqslant q^{n-\delta+1}$.

## 83. Theorem (Plotkin bound)

*Let* $n \geqslant \delta \geqslant 1$ *and write* $\theta = \frac{q-1}{q}$. *Then,* $A_q(n,\delta) \leqslant \frac{\delta}{\delta-\theta n}$.

### 2.5. Linear codes

## 84. Definition

*A code* $\mathcal{C} \subseteq A^n$ *is linear if* $q = |A|$ *is a prime (so,* $A = \mathbb{Z}/q\mathbb{Z}$ *is a field), and* $\mathcal{C}$ *is a* $\mathbb{Z}/q\mathbb{Z}$*-linear subspace of* $A^n$, *i.e.,* $\mathbf{u}, \mathbf{v} \in \mathcal{C} \Rightarrow \mathbf{u} + \mathbf{v} \in \mathcal{C}$.

## 85. Example

## 86. Definition

*The dimension of a linear code $\mathcal{C}$ is $k =$ the number of vectors in a (so, any) basis of $\mathcal{C}$. A linear $(n, M, \delta)_q$-code of dimension $k$ is also said to be an $[n, k, \delta]_q$-code or an $[n, k]_q$-code; of course, $M = |\mathcal{C}| = q^k$.*

## 87. Definition

*The weight of a vector $\mathbf{x} \in A^n$ is $|\mathbf{x}| = |\{i \mid x_i \neq 0\}|$. Clearly, $d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}|$.*

## 88. Proposition

*Let $\mathcal{C}$ be a linear code. Then $\delta(\mathcal{C}) = \min\{|\mathbf{u}| \mid \mathbf{0} \neq \mathbf{u} \in \mathcal{C}\}$.*

## 89. Explanation

*Advantages of linear codes: (i) faster computation of $\delta$; (ii) control on its $q^k$ elements by using $k$ parameters; (iii) there are good general ways to (de)codify (generating matrix), to detect (control matrix), and to correct (coset leaders); (iv) and very good specific examples (Hamming, Golay, BCH, Reed–Solomon, etc).*

## 90. Definition

*Let $\mathcal{C}$ be a $[n,k,\delta]_q$-code. A matrix $G \in \mathrm{Mat}_q(k,n)$ is a generating matrix for $\mathcal{C}$ if its rows form a basis for $\mathcal{C}$. (De)codification in $\mathcal{C}$ using a generating matrix $G$.*

## 91. Example

## 92. Definition

*Let $G \in \mathrm{Mat}_q(k,n)$. An elementary transformation, $G \sim G'$ is one of the following moves: (i) a permutation of rows; (ii) a multiplication of a row by a non-zero scalar; (iii) the addition to a row of a linear combination of the others.*

## 93. Proposition

*Let $G_1, G_2 \in \mathrm{Mat}_q(k,n)$ be matrices with L.I. rows. Then, they are generating matrices of the same linear code $\Leftrightarrow \exists$ a finite sequence of elementary transformations, $G_1 \sim \cdots \sim G_2$, passing from one to the other.*

## 94. Example

## 95. Definition

*Let $\mathcal{C}$ be a linear code. The canonical generating matrix for $\mathcal{C}$ is the one having the identity $I_k$ on its leftmost possible columns. We say that $\mathcal{C}$ is systematic if it admits a generating matrix having the identity $I_k$ on its first $k$ columns.*

## 96. Proposition

*Algorithm to find the canonical generating matrix $G_{can}$ of $\mathcal{C}$.*

## 97. Definition

*Let $\mathcal{C}$ be a linear $[n,k]_q$-code. The dual code is $\mathcal{C}^\perp = \{\mathbf{x} \mid \mathbf{u}\cdot\mathbf{x}^T = 0,\ \forall \mathbf{u} \in \mathcal{C}\}$.*

## 98. Proposition

*If $\mathcal{C}$ is a linear $[n,k]_q$-code, then $\mathcal{C}^\perp$ is a linear $[n,n-k]_q$-code.*

## 99. Corollary

*For a linear code $\mathcal{C}$, we have $(\mathcal{C}^\perp)^\perp = \mathcal{C}$. So, $\mathcal{C} = \{\mathbf{u} \mid \mathbf{u}\cdot\mathbf{x}^T = 0,\ \forall \mathbf{x} \in \mathcal{C}^\perp\}$.*

## 100. Definition

*Let $\mathcal{C}$ be linear. A control matrix for $\mathcal{C}$ is a generating matrix for $\mathcal{C}^{\perp}$. The canonical control matrix $H_{can}$ is the one having $I_k$ on the rightmost possible columns.*

## 101. Proposition

*Let $\mathcal{C}$ be a linear $[n,k]_q$-code, $G = (I_k \mid A) \in \mathrm{Mat}_q(k,n)$, and $H = (A^T \mid I_{n-k}) \in$ $\in \mathrm{Mat}_q(n-k,n)$. Then, $G$ is a gen. matrix for $\mathcal{C} \Leftrightarrow H$ is a control matrix for $\mathcal{C}$.*

## 102. Example

## 103. Proposition

*Let $\mathcal{C}$ be a linear code and $H$ a control matrix. Then, $\delta(\mathcal{C})$ is the minimum number of L.D. columns in $H$.*

### 2.7. Correction with linear codes

## 104. Definition

*Let $\mathcal{C}$ be a linear code. We can partition $A^n$ into classes modulo $\mathcal{C}$:* $\mathbf{x} \sim \mathbf{y} \Leftrightarrow \mathbf{x} - \mathbf{y} \in \mathcal{C}$. *Clearly,* $[\mathbf{x}] = \mathbf{x} + \mathcal{C}$ *and, in particular,* $[\mathbf{0}] = \mathcal{C}$.

## 105. Definition

*Let $\mathcal{C}$ be a linear $[n,k]_q$-code, and $H$ a control matrix. The syndrome of a vector $\mathbf{x} \in A^n$ is the (column) vector $s(\mathbf{x}) = H\mathbf{x}^T \in A^{n-k}$.*

## 106. Observation

*Let $\mathcal{C}$ be a linear code, and $H$ a control matrix. Then, two vectors $\mathbf{x}, \mathbf{y} \in A^n$ belong to the same class, $\mathbf{x} + \mathcal{C} = \mathbf{y} + \mathcal{C}$, if and only if $s(\mathbf{x}) = s(\mathbf{y})$.*

## 107. Definition

*A leader of a class $\mathbf{x} + \mathcal{C}$ is a vector of minimal weight on it. The leader of $\mathcal{C}$ is $\mathbf{0}$.*

## 108. Proposition

*Let $\mathcal{C}$ be a linear code with $\rho = \rho(\mathcal{C})$, let $\mathbf{y} \in A^n$, and let $\mathbf{z}$ be a leader in $\mathbf{y} + \mathcal{C}$. Then,*
*(i) $\mathbf{y} - \mathbf{z} \in \mathcal{C}$;*
*(ii) $d(\mathbf{y}, \mathcal{C}) = |\mathbf{z}| = d(\mathbf{y}, \mathbf{y} - \mathbf{z})$;*
*(iii) if $d(\mathbf{y}, \mathcal{C}) < \rho$ then $\mathbf{z}$ is the unique leader in the class $\mathbf{y} + \mathcal{C}$.*

## 109. Explanation

*Description of the general correction algorithm for linear codes.*

## 110. Example

## 2.8. The Hamming codes

## 111. Definition

*We define the binary Hamming code, $Ham_2(t)$, as that $\mathcal{C} \leqslant A^{2^t-1}$ having as a control matrix $H_t \in \mathrm{Mat}_2(t, n)$ with column $i = 1, \ldots, 2^t - 1$ reading $i$ in base 2.*

## 112. Proposition

*$Ham_2(t)$ is a linear $[n, k, \delta]_2$-code, where $n = 2^t - 1$, $k = n - t$, and $\delta = 3$. Moreover, it is perfect, optimal, and $A_2(2^t - 1, 3) = 2^{2^t-t-1}$.*

## 113. Explanation

*Fast correction with binary Hamming codes.*